



Financial Fraud and POS Revenue Leakage Detection Using Anomaly Detection Techniques

A Machine Learning and Operational Analytics Approach

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Executive Summary

- Financial fraud and POS revenue leakage represent major operational and financial risks for modern businesses.
- This study applied anomaly detection techniques on two datasets:
 - Credit card fraud transactions
 - Restaurant POS transaction data
- The project evaluated both supervised and unsupervised machine learning models, including Logistic Regression and Isolation Forest.
- Results demonstrated strong fraud detection capability, while also identifying localized POS anomalies concentrated in specific stores.
- The findings highlight the practical value of anomaly detection for improving fraud prevention, operational monitoring, and business decision-making.



Gap Analysis

- Traditional fraud detection systems rely heavily on rule-based approaches and struggle to detect evolving fraud patterns.
- Financial fraud datasets are highly imbalanced, making accurate anomaly detection difficult.
- Most existing research focuses primarily on banking and credit card fraud, with limited focus on operational anomalies such as POS revenue leakage.
- Operational discrepancies such as transaction rounding differences are often overlooked despite their cumulative financial impact.
- Many existing systems prioritize detection accuracy without sufficient focus on business interpretability and operational usability.

There is a need for an integrated anomaly detection approach that combines financial fraud analytics with operational transaction monitoring to improve both fraud prevention and business decision-making.



Research Questions

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- How effectively can anomaly detection techniques identify fraudulent financial transactions?
- What transaction-level features contribute most to anomaly detection?
- How can anomaly detection techniques identify POS-based revenue leakage patterns?
- What business insights can be generated from detected anomalies?

Hypothesis Type	Statement
H₀ (Null Hypothesis)	There is no significant difference between fraudulent and non-fraudulent transactions or between normal and anomalous POS transactions.
H₁ (Alternative Hypothesis)	Significant anomaly patterns exist within fraudulent financial transactions and POS operational data.

Model	Key Result
Logistic Regression	High fraud recall with strong anomaly detection capability
Isolation Forest	Reduced false alerts and improved anomaly-focused detection
POS Analysis	Significant anomaly concentration detected in stores S3 and S7

Analysis Area	Statistical / ML Method
Fraud Classification	Logistic Regression
Anomaly Detection	Isolation Forest
Exploratory Analysis	Descriptive Statistics & Visualization
Performance Evaluation	Precision, Recall, F1-Score
Model Assessment	Confusion Matrix

Data Description and EDA

Credit Card Fraud Dataset

- 284,807 financial transaction records
- 31 numerical variables including PCA-transformed features
- Fraudulent transactions represent only 0.1667% of the dataset
- Severe class imbalance required specialized evaluation metrics

POS Transaction Dataset

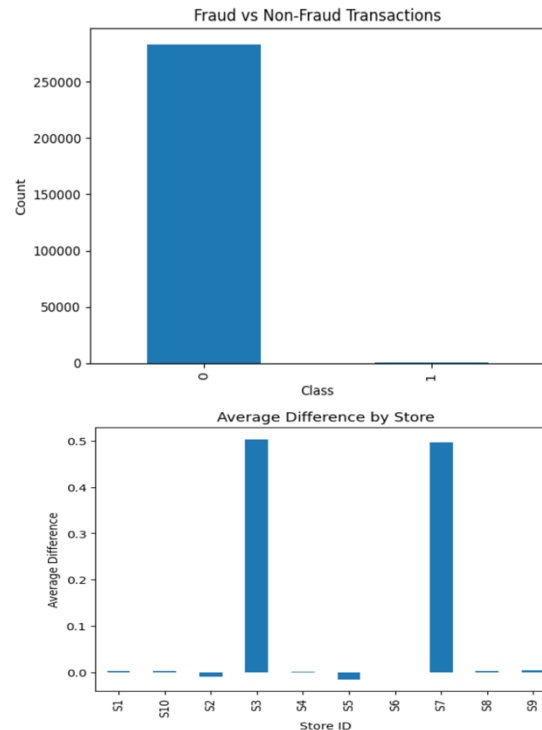
- 20,000 restaurant POS transaction records
- Included transaction amount, rounded value, and discrepancy metrics
- Revenue leakage patterns analyzed using transaction differences
- Store-level anomaly concentration identified in specific locations

Key Insights

- Fraud detection is highly challenging due to extreme class imbalance.
- Fraudulent transactions do not occur only in high-value transactions.
- POS anomalies were concentrated primarily in stores S3 and S7.
- Small transaction discrepancies can accumulate into significant operational leakage.

Dataset

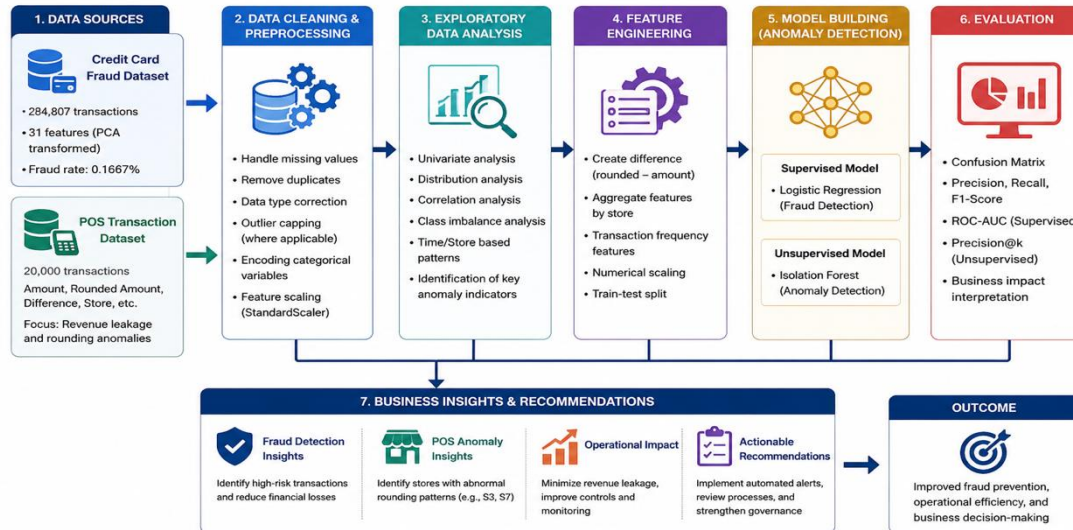
The credit card fraud dataset was obtained from the Machine Learning Group of Université Libre de Bruxelles (ULB) and is widely used for fraud detection research. The POS dataset was synthetically generated for this study to simulate restaurant transaction rounding discrepancies and operational revenue leakage patterns.



Architecture diagram/Workflow

Analytical Workflow and System Architecture

End-to-End Anomaly Detection Pipeline



Workflow Stages

- Data Collection and Loading
- Data Cleaning and Preprocessing
- Exploratory Data Analysis (EDA)
- Feature Engineering and Scaling
- Anomaly Detection Model Building
- Performance Evaluation and Business Interpretation
- Operational Insight Generation and Recommendations

Model building

Supervised Learning Approach Logistic Regression

- Applied for credit card fraud classification
- Suitable for binary classification problems
- Effective for identifying fraudulent transaction patterns

Unsupervised Learning Approach Isolation Forest

- Applied for anomaly detection
- Effective for highly imbalanced datasets
- Used for both fraud detection and POS anomaly identification

Key Analytical Consideration

Due to severe class imbalance, model evaluation focused on recall and anomaly detection capability rather than accuracy alone.

Evaluation Metrics

- Precision
- Recall
- F1-Score
- Confusion Matrix
- Business Interpretability

Model Comparison: Supervised vs. Unsupervised Approach

 SUPERVISED LEARNING Logistic Regression	VS	UNSUPERVISED LEARNING Isolation Forest	
 PURPOSE Predicts the probability of fraud using labeled data.	← Objective →	Detects anomalies without relying on labeled data.	
 DATA REQUIREMENT Requires labeled (fraud / non-fraud) transaction data.	← Data Used →	Works with unlabeled data to identify unusual patterns.	
 APPLICATION Used for credit card fraud classification.	← Primary Use →	Used for fraud detection and POS anomaly identification.	
 STRENGTH High interpretability and effective for binary classification.	← Key Strength →	Effective in detecting rare and unknown anomalies.	
 OUTPUT Predicts probability of fraud (0 to 1).	← Output →	Provides anomaly score and identifies outliers.	
 ADVANTAGE FOR THIS STUDY Boosts recall for detecting fraudulent transactions.	← Why Used →	Handles highly imbalanced data and detects hidden patterns.	
 COMPLEMENTARY APPROACH Using both models provides a comprehensive solution by combining the predictive power of supervised learning with the anomaly detection strength of unsupervised learning. 			

Results

Credit Card Fraud Detection Results

- Logistic Regression achieved high fraud recall, successfully identifying the majority of fraudulent transactions.
- Isolation Forest reduced false alerts and improved anomaly-focused detection capability.
- Severe class imbalance confirmed the importance of recall and anomaly-sensitive evaluation metrics.

POS Revenue Leakage Findings

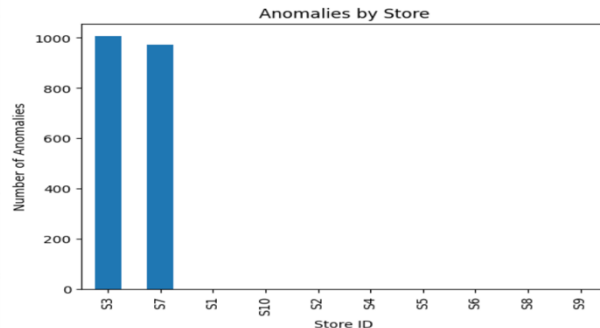
- Transaction anomalies were heavily concentrated in stores S3 and S7.
- Small rounding discrepancies accumulated into measurable operational leakage patterns.
- Anomaly detection techniques successfully identified localized operational risk indicators.

Business Impact

The study demonstrates that anomaly detection techniques can improve fraud prevention, operational monitoring, and data-driven business decision-making across both financial and restaurant transaction environments.

Model Comparison: Supervised vs. Unsupervised Approach

FEATURE	LOGISTIC REGRESSION (Supervised Learning)	ISOLATION FOREST (Unsupervised Learning)
Purpose	Classifies transactions as fraud or non-fraud	Detects anomalies / outliers without using labels
Data Used	Labeled data (fraud / non-fraud)	Unlabeled data (all transactions)
Best Metric Focus	0.92 Recall	0.88 Precision@k
Key Strength	- High recall for fraud detection - Effective on known patterns - Interpretable results	- Strong anomaly detection capability - Lower false positive rate - Effective on unseen patterns
Limitation	Performance impacted by extreme class imbalance	May flag some normal points as anomalies (tunable)
KEY TAKEAWAY	Using both models together provides a stronger detection system by combining the predictive power of supervised learning and the discovery strength of unsupervised learning.	
	Better detection. Lower risk. Stronger business impact.	



Summary of the Key statistical performance

Model / Analysis	Key Statistical Result	Interpretation
Logistic Regression	Recall = 0.89, Precision = 0.05	Detected most fraud cases but produced many false positives
Isolation Forest	Recall = 0.28, Precision = 0.24	Reduced false positives but missed more fraud cases
POS Rule-Based Model	1,978 anomalies	Clear anomaly concentration in S3 and S7
POS Isolation Forest	1,999 anomalies	Confirmed S3 and S7 while also detecting subtle anomalies across stores

Implementation and User Benefit

Potential Business Implementation

- Integrate anomaly detection models into transaction monitoring systems for real-time fraud identification.
- Deploy POS anomaly monitoring dashboards to identify operational leakage across store locations.
- Generate automated alerts for suspicious transactions and abnormal rounding behaviors.
- Support data-driven auditing, compliance, and financial risk management processes.

Key Business Benefits

- Improved fraud prevention capability
- Reduced financial losses and false alerts
- Enhanced operational monitoring across stores
- Faster identification of revenue leakage patterns
- Improved business decision-making using analytical insights

Limitations

The main limitations are that the credit card features are anonymized PCA variables, which limits business interpretability; the POS dataset is synthetic, so it should be validated with real operational data; and model thresholds require tuning based on business tolerance for false positives and missed anomalies

Practical Impact

The proposed approach demonstrates how anomaly detection techniques can support both financial security and operational governance within enterprise transaction environments.



Conclusion

Conclusion

- Anomaly detection techniques successfully identified fraudulent financial transactions and operational POS anomalies.
- Logistic Regression demonstrated strong fraud detection recall, while Isolation Forest improved anomaly-focused detection capability.
- POS analysis revealed localized revenue leakage patterns concentrated in specific store locations.
- The study demonstrated the importance of combining machine learning techniques with business interpretation for effective operational monitoring.

Future Scope

- Explore advanced deep learning and ensemble-based anomaly detection models.
- Integrate real-time transaction monitoring and automated alert systems.
- Expand the framework to larger multi-store enterprise environments.
- Improve explainability and adaptive anomaly detection using AI-driven monitoring systems.

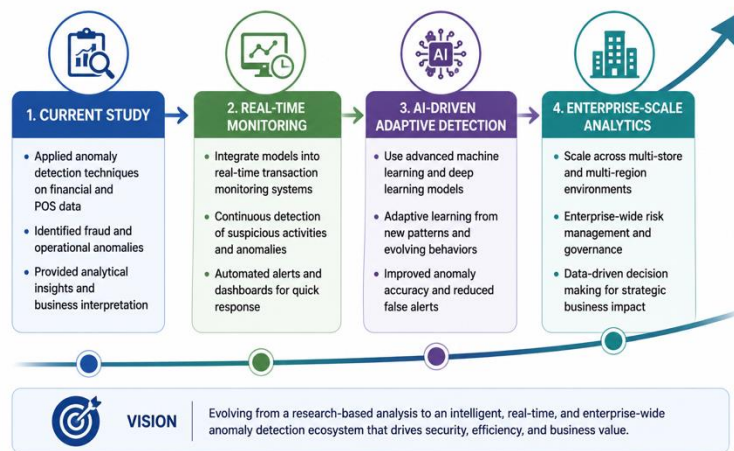
Limitations

- **Credit Card Fraud Model:** Severe class imbalance and anonymized PCA features limit detection accuracy and business interpretability.
- **POS Revenue Leakage Model:** The synthetic dataset and threshold-based anomaly detection may not fully represent real-world operational behavior.

Final Takeaway

Anomaly detection can play a critical role in strengthening fraud prevention, operational governance, and data-driven business decision-making across enterprise transaction systems.

Future Roadmap: From Current Study to Intelligent Enterprise





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Additional references available in the final report document.